

JEE MAINS GRAND TEST SERIES



NARAYANA
IIT ACADEMY
INDIA

40+
YEARS
OF EXCELLENCE

Sec: SR.IIT_*CO-SC(MODEL-A,B&C)

Time: 3HRS

Date:13-01-24

Max. Marks: 300

Name of the Student: _____

H.T. NO:

13-01-24_SR.STAR CO-SUPER CHAINA(MODEL-A,B&C)_JEE-MAIN_GTM-15(N)_SYLLABUS

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

MATHEMATICS: **TOTAL SYLLABUS**

SUBJECT	MISTAKES		
	JEE SYLLABUS Q'S	JEE EXTRA SYLLABUS Q'S	TOTAL Q'S
MATHS			
PHYISCS			
CHEMISTRY			

For More Material Join: @JEEAdvanced_2024

PHYSICS

MAX.MARKS: 100

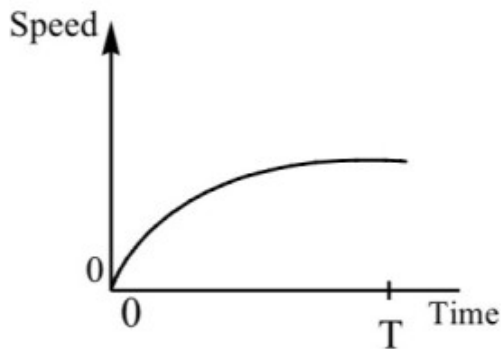
SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

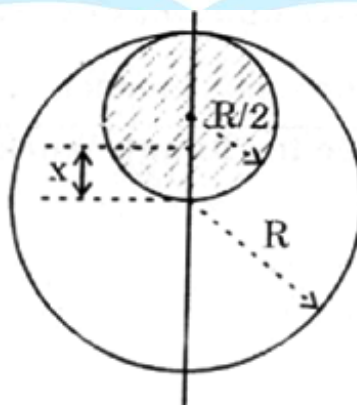
1. If 'P' represents radiant energy per unit volume, 'C' speed of light and 'Q' radiant energy striking per unit area per second, then nonzero integers x, y and z such that $P^x Q^y C^z$ is dimensionless, may be
- A) $x=1, y=1, z=1$ B) $x=1, y=-1, z=1$ C) $x=-1, y=1, z=1$ D) $x=1, y=1, z=-1$
2. A stone is dropped from the top of a tall cliff and n seconds later another stone is thrown vertically downwards with a velocity u. Then the second stone overtakes the first, below the top of the cliff at a distance given by
- A) $\frac{g}{2} \left[\frac{n \left(\frac{gn}{2} - u \right)}{(gn - u)} \right]^2$ B) $\frac{g}{2} \left[\frac{n(gn - u/2)}{(gn - u)} \right]^2$ C) $g \left[\frac{(gn - u)}{(gn - u/2)} \right]^2$ D) None
3. In a screw gauge, there are 100 divisions in circular scale and each main scale division is of 1 mm. When there is no gap between the jaws, 97th division coincides with the main scale zero and zero of main scale is not visible. While measuring the diameter of a ball, the circular scale is between 3 mm mark and 4 mm mark such that the 76th division of circular scale coincides with the reference line. Select the correct alternative
- A) the least count of the micrometer is 0.01 cm
B) the zero error is - 0.04 mm
C) the diameter of the ball is 3.79 mm
D) the main scale reading is 4 mm
4. A block of mass m is placed on a smooth wedge of inclination θ . The whole system is accelerated horizontally so that the block does not slip on the wedge. The force exerted on the block by the wedge has a magnitude.
- A) mg B) $\frac{mg}{\cos \theta}$ C) $mg \cos \theta$ D) $mg \tan \theta$

5. The variation of the speed with time, of a ball falling in air is shown in below.



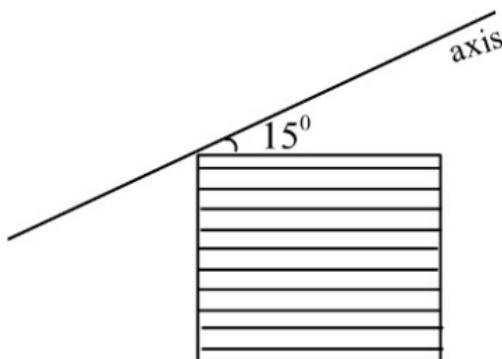
During the time from 0 to T, the ball gains some kinetic energy and loses gravitational potential energy ΔE_p . Which of the following statements must be correct?

- A) ΔE_p is equal to the gain in kinetic energy
B) ΔE_p is equal to the work done against air resistance
C) ΔE_p is greater than the gain in kinetic energy
D) ΔE_p is less than the work done against air resistance
6. A sphere of styrofoam has radius R. A cavity of radius $R/2$ having center at a distance $R/2$ directly above the center of the sphere is hollowed out and filled with a solid material of density five times the density of Styrofoam (figure). Where is this center of mass of the new sphere? (From the center of the sphere)

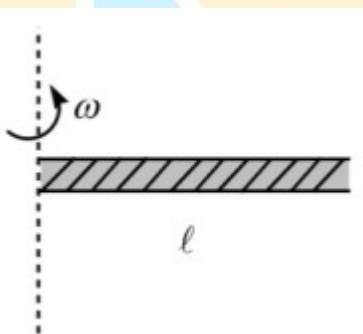


- A) $\frac{R}{3}$ B) $\frac{R}{4}$ C) $\frac{R}{6}$ D) $\frac{3R}{16}$

7. A uniform square plate of mass M side length ' L ' as shown in fig. Then the moment of inertia of the square plate about an axis makes an angle 15° with one side and present in the same plane.



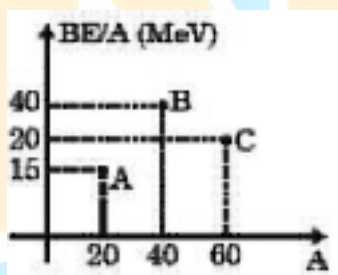
- A) $\frac{ML^2}{6}$ B) $\frac{ML^2}{12}$ C) $\frac{11ML^2}{24}$ D) $\frac{7ML^2}{12}$
8. A thin horizontal tube of inner radius ' r ' and length ' ℓ ' is open at both ends and is fully filled with a liquid of density ρ and surface tension ' σ '. If the rod is rotated about a vertical axis passing through one of its end perpendicular to its length at a constant angular velocity ω . Find maximum value of ω for which the liquid doesn't spill out of the tube:



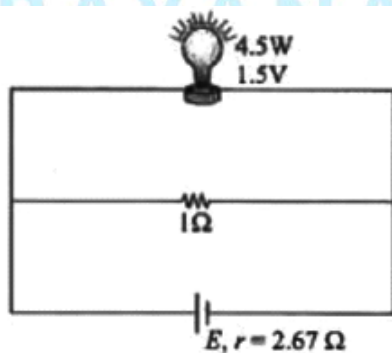
- A) $\sqrt{\frac{8\sigma}{\rho r \ell^2}}$ B) $\sqrt{\frac{6\sigma}{\rho r \ell^2}}$ C) $\sqrt{\frac{4\sigma}{\rho r \ell^2}}$ D) $\sqrt{\frac{2\sigma}{\rho r \ell^2}}$
9. A piece of metal floats on mercury. The cubic expansivities of the metal and mercury are γ_1 and γ_2 respectively. If the temperature is increased by ΔT , the fraction of the volume of the metal submerged in mercury changes by a factor of (Assume $\gamma \Delta T \ll 1$)

- A) $\frac{1+\gamma_1\Delta T}{1+\gamma_2\Delta T}$ B) $\frac{1+\gamma_2\Delta T}{1+\gamma_1\Delta T}$ C) $\frac{1-\gamma_2\Delta T}{1-\gamma_1\Delta T}$ D) $\frac{1+\gamma_1\Delta T}{1-\gamma_2\Delta T}$

10. A flat horizontal board moves up and down in SHM of amplitude A . Then the shortest permissible time period of the vibration such that an object placed on the board may not lose contact with the board is
- A) $2\pi\sqrt{\frac{g}{A}}$ B) $2\pi\sqrt{\frac{A}{g}}$ C) $2\pi\sqrt{g \times A}$ D) $\frac{1}{2\pi}\sqrt{\frac{g}{A}}$
11. A student performs an experiment for determination of $g = \left(\frac{4\pi^2\ell}{T^2}\right)$, $\ell = 1\text{m}$ and he commits an error of $\Delta\ell$. For T he takes the time of n oscillation with the stop watch of least count Δt and he commits a human error of 0.1 sec. For which of the following data, the measurement of g will be most accurate?
- A) $\Delta L = 0.5, \Delta t = 0.1, n = 20$ B) $\Delta L = 0.5, \Delta t = 0.1, n = 50$
 C) $\Delta L = 0.5, \Delta t = 0.01, n = 20$ D) $\Delta L = 0.5, \Delta t = 0.05, n = 50$
12. The graph shows BE/A values for three hypothetical elements A, B, C. The energy released when C disintegrates into A and B is :



- A) 1000 MeV B) 1200 MeV C) 700 MeV D) 500 MeV
13. A torch bulb rated as 4.5 W, 1.5 V is connected as shown in fig. The e.m.f of the cell needed to make the bulb glow at full intensity is

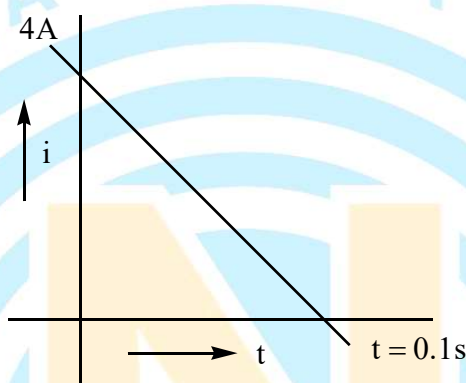


- A) 4.5 V B) 1.5 V C) 2.67 V D) 13.5 V

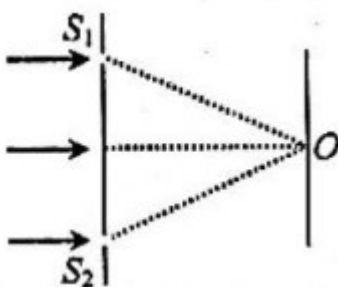
14. At $t = 0$, a charged particle of specific charge α moving with a velocity $(V\hat{i} + V\sqrt{3}\hat{k})$ enters a region of uniform electric and magnetic fields given by $E\hat{j}$ and $-B\hat{j}$ respectively. Speed of that particle becomes $3V$ at time $t =$

- A) $\frac{\sqrt{2}V}{\alpha E}$ B) $\frac{2V}{\alpha E}$ C) $\frac{\sqrt{5}V}{\alpha E}$ D) $\frac{\sqrt{3}V}{\alpha E}$

15. Some magnetic flux is changed from a coil of resistance 10Ω . As a result an induced current is developed in it which varies with time as shown in figure. The magnitude of change in flux through the coil in weber is :-

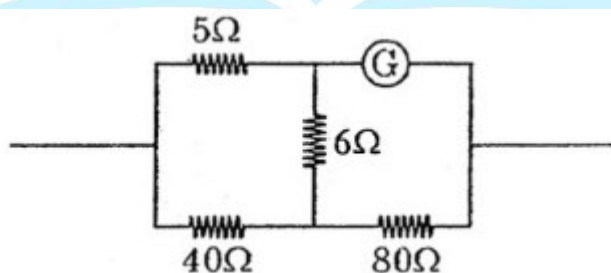


- A) 2 B) 4 C) 6 D) None of these
16. A plane electromagnetic wave travelling in a non-magnetic medium is given by $E = (9 \times 10^8 \text{ NC}^{-1}) \sin[(9 \times 10^8 \text{ rads}^{-1})t - (6 \text{ m}^{-1})x]$, where x is in metre and t is in second. The dielectric constant of the medium is
- A) 5 B) 4 C) 3 D) 2
17. Statement-1 : In YDSE, as shown in figure, central bright fringe is formed at O. If a liquid is filled between plane of slits and screen, the central bright fringe is shifted in upward direction.



Statement – 2 : If path difference at O increases y-coordinate of central bright fringe will change.

- A) Statement -1 is true, statement-2 is true and statement-2 is correct explanation for statement-1
- B) Statement -1 is true, statement-2 is true and statement-2 is not correct explanation for statement-1
- C) Statement-1 is true, statement-2 is false
- D) Statement-1 is false, statement-2 is true
18. When a metal plate is exposed to light of wavelength 400nm, a negative potential of 1.1v is found to be needed to stop the photo current. The threshold wavelength of the photo electrons is
- A) 450 nm B) 500 nm C) 580 nm D) 620 nm
19. An α - particle and a proton are accelerated from rest by a potential difference of 100V. After this their de-Broglie wave lengths are λ_α and λ_p respectively. The ratio $\frac{\lambda_p}{\lambda_\alpha}$ is
- A) $\frac{2\sqrt{2}}{1}$ B) $\frac{1}{2\sqrt{2}}$ C) $\frac{\sqrt{2}}{1}$ D) $\frac{1}{\sqrt{2}}$
20. A galvanometer of $i_{g\max} = 1\text{mA}$ & resistance = 10Ω is connected in a circuit as shown. This combination (circuit) can be used as an ammeter or a voltmeter also. The range of ammeter (R_A) and voltmeter (R_V) are :



- A) $\frac{9}{8}\text{mA}$, 15mV B) 1mA, 12mV C) $\frac{8}{3}\text{mA}$, 20mV D) $\frac{4}{3}\text{mA}$, 10mV

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

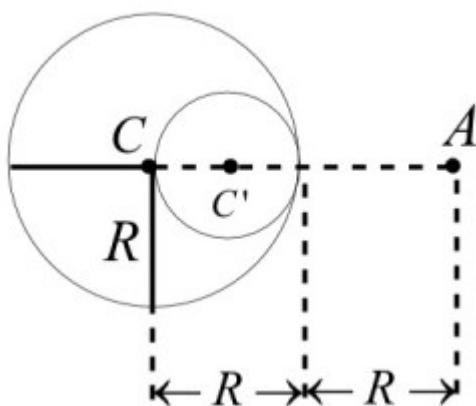
Marking scheme: +4 for correct answer, -1 in all other cases.

21. Gravitational potential at A, due to a uniform solid sphere of mass M and radius R is V_1 .

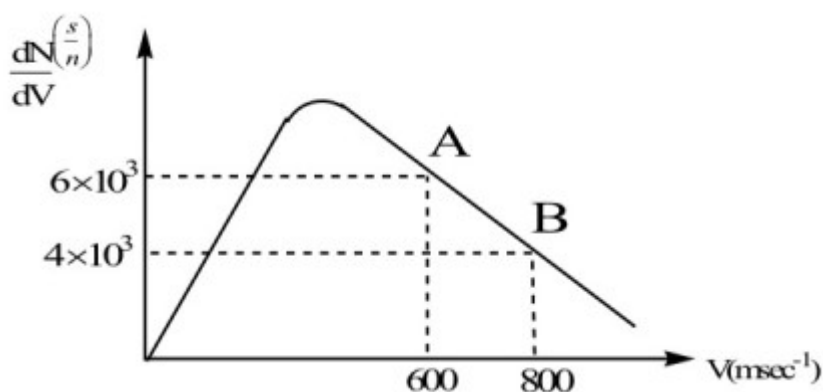
If a spherical part of radius $\frac{R}{2}$ is removed from the sphere as shown in figure, the

gravitational potential at A due to remaining part becomes V_2 . Find the value of

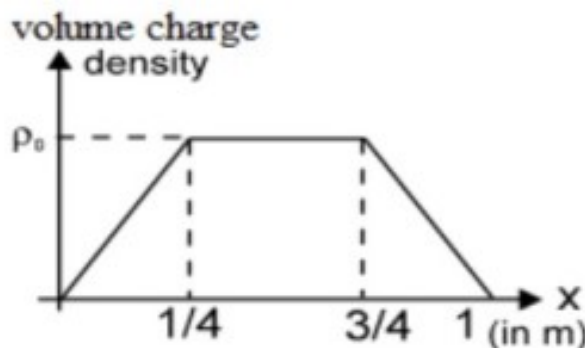
$$(V_1 / V_2) - 0.2$$



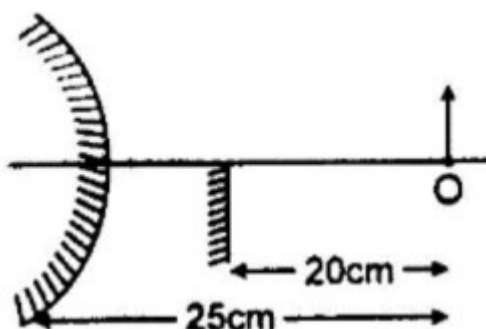
22. A liquid is kept in a cylindrical vessel which is rotated along its axis. The liquid rises at the side. If the radius of the vessel is 0.05m and the speed of rotation is 2 rev/sec. Find the difference in the height of the liquid at the centre of the vessel and its sides (in cm). Take $g = 10 \text{ m/s}^2$ & $\pi^2 = 10$
23. Maxwells speed distribution curve for an ideal gas is shown in the figure. If the number of the molecules in the velocity range from 600 m/s and 800 m/s is 10^n , find the value of n. Assume that the graph between the points A and B is linear.



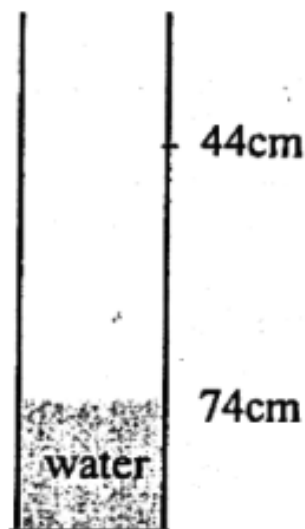
24. The volume charge density as a function of distance 'x' from one face inside a cube of side length 'a' is varying as shown in the figure. Then the total flux in SI units through the cube is $\frac{na^2\rho_0}{4\epsilon_0}$. Find n.



25. A capacitor of capacitance $1\mu\text{F}$ can with stand a maximum voltage of 6 kV and the other capacitor of capacitance $2\mu\text{F}$ can with stand a maximum voltage of 4 kV. The series combination of the capacitors can with stand a maximum voltage of n kV. What is n?
26. An iron rod of volume 10^{-4} m^3 and relative permeability 1000 is placed inside a long solenoid wound with 5 turns/cm. If a current of 0.5A is passed through the solenoid, then the magnetic moment of the rod is (in $\text{A} - \text{m}^2$)
27. In the figure, an object is placed 25cm from the surface of a convex mirror, and a plane mirror is set so that the image formed by the two mirrors lie adjacent to each other in the same plane. The plane mirror is placed at 20cm from the object. The radius of curvature of the convex mirror (in cm) is $k \times 25$, then the value of k is _____



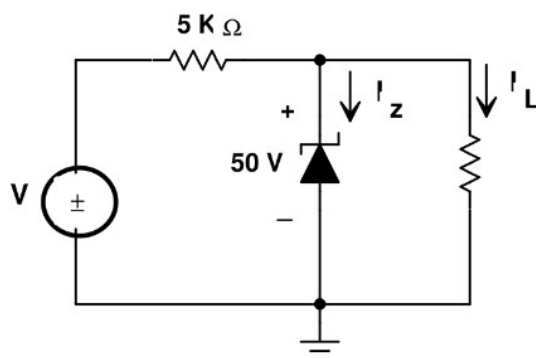
28. Sound waves of frequency 320Hz are sent into the top of a vertical tube containing water at a level that can be adjusted. Standing waves are produced at two any suitable successive water levels 44cm and 74cm from open end. The distance of nearest displacement antinode from open end (in cm) is



29. An electron in hydrogen atom jumps from n_1 state to n_2 state, where n_1 and n_2 represent the quantum number to two states. The time period of revolution of electron in the initial state is 8 times than in final state. Then the ratio of n_1 and n_2 is
30. In the adjacent figure a zener diode acts as a voltage regulator the current through the load I_L and the current through the zener diode I_z are given below, then find the difference of maximum and minimum voltage of source (in volts).

$$0 \leq I_L \leq 4 \text{ mA}$$

$$2 \leq I_z \leq 8 \text{ mA}$$

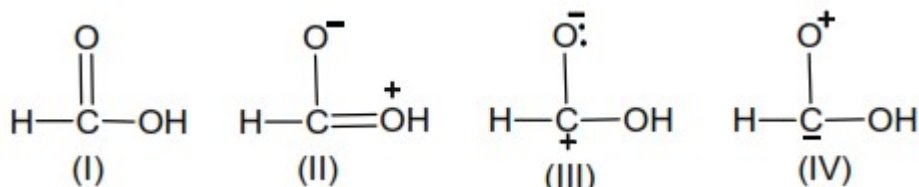


SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

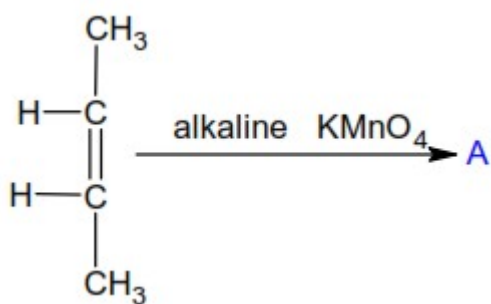
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

31. Increasing order of stability is



- A) I < III < II < IV B) IV < III < II < I
C) III < IV < II < I D) II < IV < III < I

32. Which is true about this reaction?



- A) A is meso 1, 2-butan-di-ol formed by syn addition
B) A is meso 1, 2-butan-di-ol formed by anti addition
C) A is racemic mixture of d and ℓ , 1, 2-butan-di-ol formed by anti addition
D) A is racemic mixture of d and ℓ , 1, 2-butan-di-ol formed by syn addition

33. pH of a saturated solution of silver salt of monobasic acid HA is found to be 9. Select the option(s) which is / are correct. Given : $K_a(\text{HA}) = 10^{-10}$

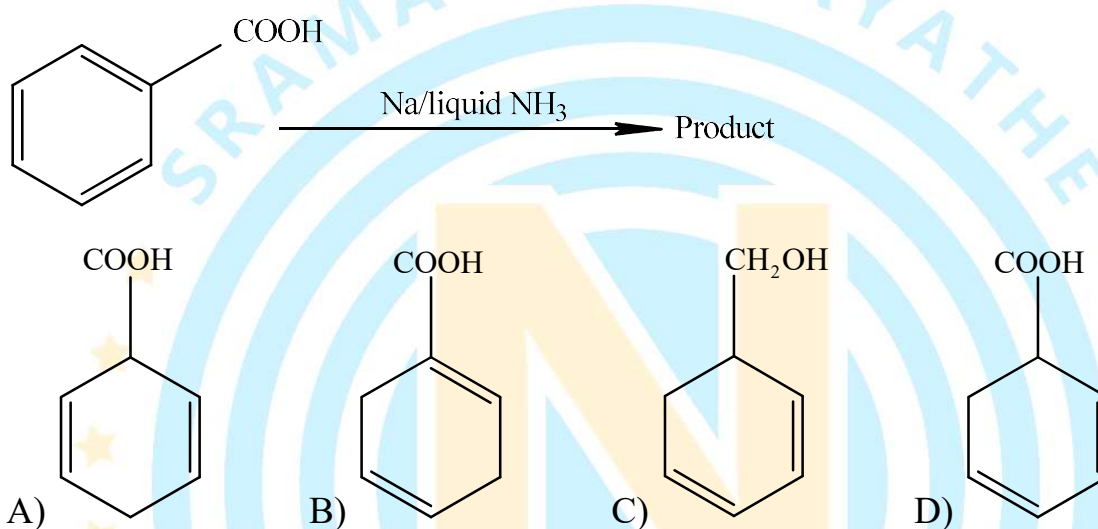
- I) Solubility of the silver salt is 1.1×10^{-5} mol / lit
II) Solubility product of the silver salt is $1.1 \times 10^{-11} \text{ M}^2$
III) Concentration of Ag^+ in the solution will be greater than that of A^-

- A) Only I, II B) Only II, III C) Only I, III D) I, II & III

34. When photons of energy 4.25 eV strike the surface of a metal A, the ejected photoelectrons have maximum kinetic energy, T_A (expressed in eV) and de-Broglie wavelength λ_A . The maximum kinetic energy of photoelectrons liberated from another metal B by photons of energy 4.20 eV is $T_B = (T_A - 1.50) \text{ eV}$. If the de-Broglie wavelength of these photoelectrons is $\lambda_B = 2\lambda_A$, then

- A) the work function of A is 2.05 eV B) the work function of B is 3.20 eV
C) $T_A = 2.00 \text{ eV}$ D) $T_B = 2.75 \text{ eV}$

35. The major product formed in the following reaction is



36. Which of the following is most reactive towards SN^2 reaction?

- A) $\text{H}_2\text{C}=\text{CH}-\text{CH}_2-\text{Cl}$ B) $\text{Ph}-\text{CH}_2-\text{Cl}$
C) $\text{Me}-\text{O}-\text{CH}_2-\text{Cl}$ D) $\text{Ph}-\text{C}(=\text{O})-\text{CH}_2-\text{Cl}$

37. The heat capacities of A, B and C are in the ratio 1 : 2 : 3. The enthalpy change for the reaction, $\text{A} + \text{B} \rightarrow \text{C}$, at temperature T_1 is ΔH_1 . Assuming the heat capacities do not change with temperature the enthalpy change, ΔH_2 at temperature T_2 will be

- A) greater than ΔH_1
B) equal to ΔH_1
C) less than ΔH_1
D) greater than (or) less than ΔH_1 depending on whether T_2 is greater (or) less than T_1

38. Given $E_{\text{Cr}^{3+}/\text{Cr}}^0 = -0.72\text{V}$, $E_{\text{Fe}^{2+}/\text{Fe}}^0 = -0.42\text{V}$.

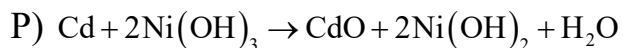
The potential for the cell $\text{Cr}|\text{Cr}^{3+}(0.1\text{M})||\text{Fe}^{2+}(0.01\text{M})|\text{Fe}$ is _____

- A) 0.339 V B) - 0.339 V C) - 0.26 V D) 0.26 V

39. Match List –I and List – II

List – I

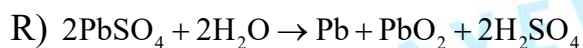
List – II



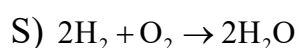
i) Primary battery



ii) Discharge of secondary battery



iii) Fuel cell



iv) Charging of secondary battery

P	Q	R	S
A) I	II	III	IV
B) IV	I	II	III
C) II	I	IV	III
D) II	I	III	IV

40. Which of the following statement(s) is/are true?

- i) In N_2 , the doubly degenerate π_{2p} orbitals are completely filled.
 ii) In O_2 , the energy of σ_{2p_z} orbital is lower than the doubly degenerate π_{2p} orbitals
 iii) Different molecular species with the same configuration have the same energy.
 iv) A π_{2p}^* orbital has two nodal planes

- A) i, ii and iv B) i and ii only C) i, ii, iii and iv D) ii, iii, iv

41. What is invert sugar and why is it so named?

- A) The sugar mixture from hydrolysis of sucrose; fructose is isomerized to glucose
 B) The sugar mixture from hydrolysis of sucrose; the optical rotation changes from (+) to (-).
 C) The sugar mixture from hydrolysis of starch; on alpha glycosidic bonds are changed to beta glycosidic bonds
 D) The sugar mixture from hydrolysis of starch; glucose is isomerized to fructose.

42. $\text{CrO}_4^{2-} \xrightleftharpoons[\text{pH=Y}]{\text{pH=X}} \text{Cr}_2\text{O}_7^{2-}$ the approximate pH values X and Y are respectively.

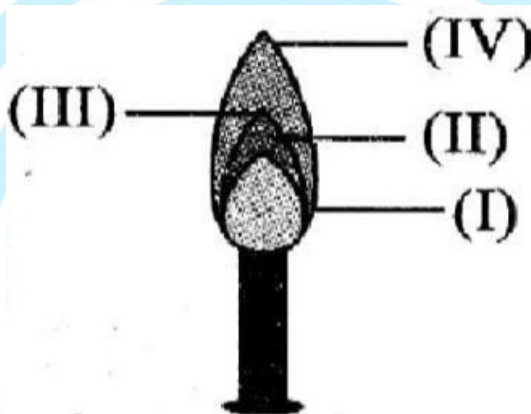
- A) 8 and 10 B) 8 and 4 C) 4 and 8 D) 6 and 3

43. Which of the following system has maximum number of unpaired electrons?

- A) d^4 (octahedral, low spin) B) d^6 (tetrahedral)
C) d^6 (Octahedral, low spin) D) d^9 (octahedral)

44. **Statement – I:** A six coordinate complex of formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$ has green colour. One litre of 0.1M solution of the complex when treated with excess of AgNO_3 gave 28.7 g of white precipitate. The formula of the complex would be: $[\text{CrCl}(\text{H}_2\text{O})_5]\text{Cl}_2 \cdot \text{H}_2\text{O}$

Statement – II: One is lowest temperature zone in Bunsen flame?



Which of the following are correct?

- A) Both statements in correct
B) Both statements are correct
C) Statement I is correct but statement II is wrong
D) Statement I is wrong but statement II is correct

45. Hydrogen bonding is absent in following compound

- A) K_2HPO_3 B) CH_3COOH C) KHCO_3 D) Urea

46. The correct order of electron gain enthalpy is:

- A) $\text{O} > \text{S} > \text{Se} > \text{Te}$ B) $\text{Te} > \text{Se} > \text{S} > \text{O}$ C) $\text{S} > \text{O} > \text{Se} > \text{Te}$ D) $\text{S} > \text{Se} > \text{Te} > \text{O}$

47. Identify the incorrect statement :

- A) The S-S-S bond angles in the S_8 and S_6 rings are the same.
- B) Rhombic and monoclinic sulphur have S_8 molecules
- C) S_2 is paramagnetic like oxygen
- D) S_8 ring has a crown shape

48. Glycerol is separated in soap industries by :

- A) Fractional distillation
- B) Differential extraction
- C) Steam distillation
- D) Distillation under reduced pressure

49. In which ion, number of unpaired electrons are not equal to four?

- (A) Fe^{2+} (B) Cr^{2+} (C) Mn^{3+} (D) Co^{2+}

50. Ammonia forms the complex $[Cu(NH_3)_4]^{2+}$ with copper ions in alkaline solution but not in acidic solution. The reason for this is:

- A) In alkaline solution $Cu(OH)_2$ is precipitated which is soluble in excess of alkali
- B) Copper hydroxide is amphoteric substance
- C) In acidic solution hydration protects Cu^{2+} ions
- D) In acidic solution protons are coordinated with ammonia molecules forming NH_4^+ ions

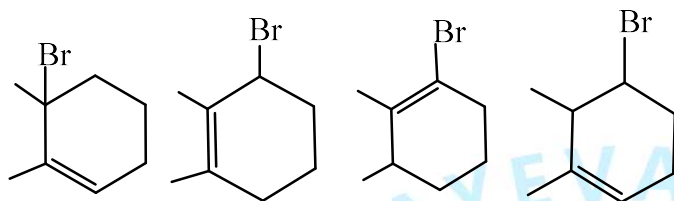
SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

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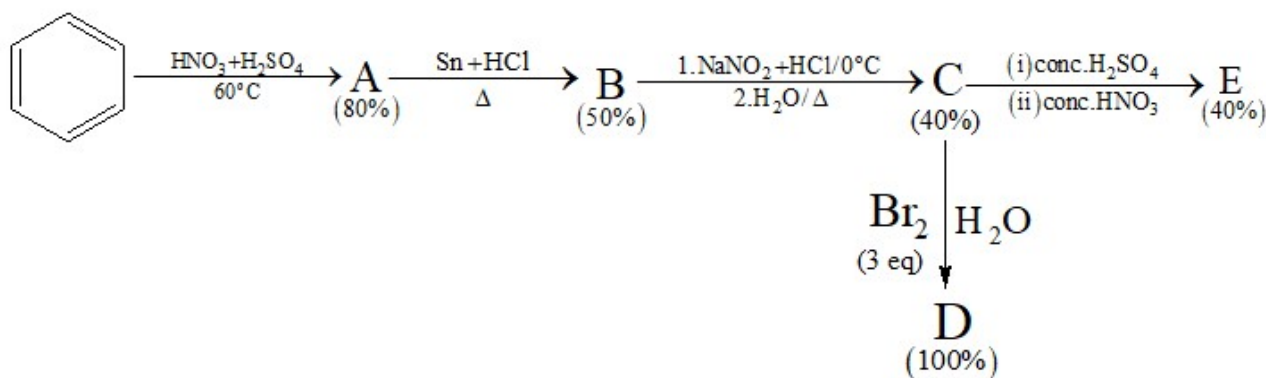
Marking scheme: +4 for correct answer, -1 in all other cases.

51. What is the sum of position number assigned to Bromine atoms while numbering the parent chain in given structures :



52. In the following reaction sequence, the % yield corresponding to the product in each step is given in the parenthesis.

(A.W (in g.mol^{-1}): H = 1, C = 12, N = 14, O = 16 & Br = 80)



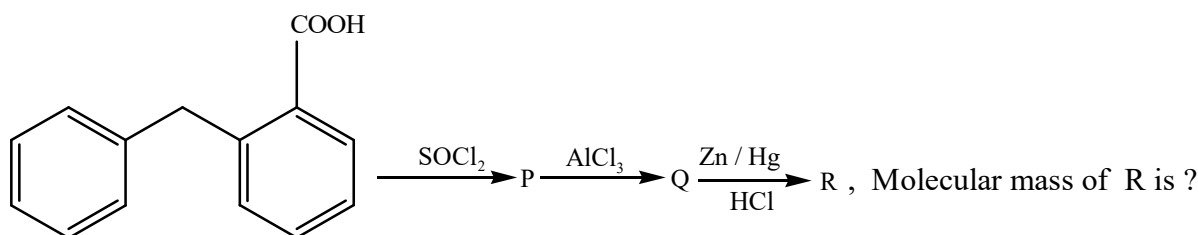
The amount of D (in g) formed from 10 moles of benzene is

53. A non volatile solid P is dissolved in liquid X and progress of first order decomposition P (in solution phase) $\rightarrow$ X(l) + 2Q(g) is monitored by total pressure measurement at 27°C . Data collected is:

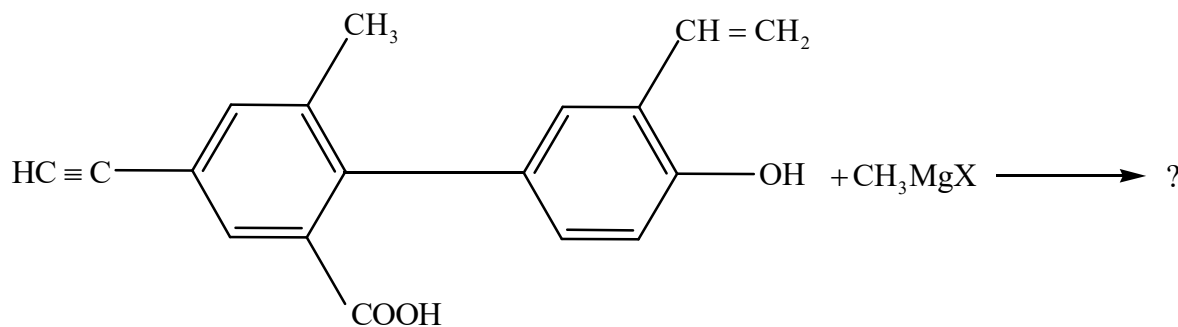
Time	$t = 0$	$t = 10 \text{ min}$	$t = \text{after long time}$
Total pressure	20 mm Hg	80 mm Hg	100 mm Hg

Calculate Half life of decomposition in minute.

- 54.



55.



How many mole alkane will be formed?

56. At 25°C vapor pressure of volatile liquid 'A' in pure state is 60 torr. To 4 mole of 'A', 'X' mole of 'B' volatile liquid is added, at 25°C . The vapour pressure of resulting solution was found to be 72 torr and mole fraction of 'A' in vapor phase is $\frac{1}{3}$, the value of 'X' is

57. With t taken in seconds and I taken in amp, the variation of I follows the equation $t^2 + I^2 = 25$. What amount of Ag will be electrodeposited with this current flowing in the interval 0-5 seconds?

(At. Mass of Ag = 108)

58. A vessel contains A(g) and B(g) at 2 atm and 4 atm respectively at TK, the mixture is allowed to attain equilibrium at TK, according to the reaction



At equilibrium, $\left(\frac{n_A}{n_B}\right)_{\text{eq}} = \left(\frac{n_B}{n_A}\right)_{\text{initial}}$

Find the value of y if $K_c = 2^y$

59. In the redox reaction, $x\text{NO}_3^- + y\text{As}_2\text{S}_3 + z\text{H}_2\text{O} \rightarrow \dots\text{AsO}_4^{3-} + \dots\text{NO} + \dots\text{SO}_4^{2-} + \dots\text{H}^+$. What is the value of $\frac{x}{z}$?

60. For the reaction, $\text{N}_2\text{O}_5(\text{g}) \rightleftharpoons 2\text{NO}_2(\text{g}) + 0.5\text{O}_2(\text{g})$. Assume ideal behavior, the mole fraction of N_2O_5 decomposed at a constant volume and temperature is $(x \times 0.1)$, if the initial pressure is 600 mm Hg and the pressure at any time is 960 mm Hg. Find the value of x .

MATHEMATICS**MAX.MARKS: 100****SECTION – I****(SINGLE CORRECT ANSWER TYPE)**

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

61. Consider a square with vertices at $(1,1)$, $(-1,1)$, $(-1,-1)$ and $(1,-1)$. Let S be the region consisting of all points inside the square which are nearer to the origin than any side, then the area of the region 'S' is
- A) $\frac{1}{12}(18\sqrt{2} - 20)$ B) $\frac{1}{12}(16\sqrt{2} - 20)$ C) $\frac{1}{3}(18\sqrt{2} - 20)$ D) $\frac{1}{3}(16\sqrt{2} - 20)$
62. For $n \in \mathbb{N}$ and $x \in [1, \infty)$, let $f_n(x) = \int \left(x^2 + (\cos \theta) \sqrt{x^2 - 1} \right)^n d\theta$. Then which one of the following is true?
- A) $f_n(x)$ is not polynomial in x if n is odd and $n \geq 3$.
- B) $f_n(x)$ is not polynomial in x if n is even and $n \geq 4$.
- C) $f_n(x)$ is a polynomial in x for all $n \in \mathbb{N}$.
- D) $f_n(x)$ is not a polynomial in x for any $n \geq 3$.
63. The number of real numbers 'a' for which the quadratic equation $x^2 + ax + 6a = 0$ has only distinct integer roots for 'x' is
- A) 8 B) 10 C) 12 D) 5
64. A batsman can score 0,2,3,4 runs for each ball he hits. The number of ways of scoring a total of 20 runs in an ordinary over (wide balls, no balls are avoided and an over contains 6 balls), is
- A) 96 B) 90 C) 315 D) 120
65. If the probability that a number selected from the set $\{1,2,3,\dots,1000\}$ is divisible by 3 but neither divisible by 5 nor by 7 is $\frac{m}{n}$, (where m and n are relatively prime numbers) then $(n-m)$ is
- A) 667 B) 717 C) 771 D) 766

66. In a test student either guesses or copies or knows the answer to a multiple choice questions with four choices in which exactly one choice is correct. The probability that he makes a guess is $\frac{1}{3}$; The probability that he copies the answer is $\frac{1}{6}$. The Probability that his answer is correct given that he copied it is $\frac{1}{8}$. Find the probability that he knew the answer to the question given that he correctly answered it is
- A) $\frac{29}{35}$ B) $\frac{24}{29}$ C) $\frac{1}{7}$ D) $\frac{1}{9}$
67. The least perimeter of an isosceles triangle in which a circle of radius 'r' can be inscribed is
- A) $4\sqrt{3}r$ B) $2\sqrt{3}r$ C) $6\sqrt{3}r$ D) $8\sqrt{3}r$
68. Let $f : [0, \infty) \rightarrow [0, \infty)$, $g : [0, \infty) \rightarrow [0, \infty)$ be two onto functions and $f(x)$ is an increasing function and $g(x)$ is a decreasing function. Also, $h(x) = f(g(x))$, $h(0) = 0$ and $p(x) = h(x^3 - 2x^2 + 2x) - h(4)$ then for every $x \in (0, 2]$
- A) $p(x) \in [0, -h(4))$ B) $p(x) \in [-h(4), 0]$
C) $p(x) \in (-h(4), h(4))$ D) $p(x) \in (h(4), -h(4)]$
69. Consider the lines L_1 and L_2 given by $L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$ and $L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$. A line having direction ratios 1, -1, -2, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is
- A) $2\sqrt{6}$ B) $3\sqrt{2}$ C) $4\sqrt{3}$ D) 4
70. The range of λ for which the line $y = x + \lambda$ passes through the common region of the circles $x^2 + y^2 - 2x - 4y + 4 = 0$ & $4(x^2 + y^2) - 16y - 20x + 37 = 0$ is
- A) $\left(\frac{1}{4} - \frac{\sqrt{7}}{4}, \frac{1}{4} + \frac{\sqrt{7}}{4}\right)$ B) $\left(-\frac{\sqrt{7}}{4}, \frac{\sqrt{7}}{4}\right)$
C) $\left(\frac{1}{2} - \frac{\sqrt{7}}{4}, \frac{1}{2} + \frac{\sqrt{7}}{4}\right)$ D) $\left(1 - \frac{\sqrt{7}}{4}, 1 + \frac{\sqrt{7}}{4}\right)$

71. If $P(h, k)$ be point on the parabola $x = 4y^2$, which is nearest to the point $Q(0, 33)$, then the distance of P from the directrix of the parabola $y^2 = 4(x + y)$ is equal to:
- A) 2 B) 4 C) 8 D) 6
72. Let R_1 and R_2 be relations on the set $\{1, 2, \dots, 50\}$ such that
- $$R_1 = \{(P, P^n) : P \text{ is a prime number and } n \geq 0 \text{ is an integer}\}$$
- $$R_2 = \{(P, P^n) : P \text{ is a prime number and } n = 0 \text{ or } 1\}$$
- Then, the number of elements in $R_1 - R_2$ is _____.
- A) 7 B) 49 C) 8 D) 51
73. If α, β, γ and δ are in A.P and $\int_0^2 f(x) dx = -4$, where $f(x) = \begin{vmatrix} x+\alpha & x+\beta & x+\alpha-\gamma \\ x+\beta & x+\gamma & x-1 \\ x+\gamma & x+\delta & x-\beta+\delta \end{vmatrix}$, then common difference of given A.P is
- A) 1 B) 0 C) 2 D) -2
74. Area bounded by the curve $a^2y = x^2(x+a)$ and X-axis is
- A) $\frac{a^2}{3}$ B) $\frac{a^2}{4}$ C) $\frac{a^2}{8}$ D) $\frac{a^2}{12}$
75. The solution of the differential equation $\frac{dy}{dx} = e^{y+x} + e^{y-x}$ is
- A) $e^{-y} = e^{-x} - e^x + c$ B) $e^{-y} = e^x - e^{-x} + c$
C) $e^{-y} = e^{-x} + e^x + c$ D) $e^y = e^x - e^{-x} + c$
76. If 'K' is a positive integer such that $36+K, 300+K, 596+K$ are the squares of three consecutive terms of an arithmetic progression, then
- Statement 1: There are 2 such progressions.
Statement 2: Common difference of A.P is ± 4
- A) Both the statements are true
B) Both the statements are false
C) Statement 1 is true and statement 2 is false
D) Statement 1 is false and statement 2 is true.

77. If the co-efficient of x^9 in $\left(\alpha x^3 + \frac{1}{\beta x}\right)^{11}$ and the co-efficient of x^{-9} in $\left(\alpha x - \frac{1}{\beta x^3}\right)^{11}$ are equal, then $(\alpha\beta)^2$ is equal to ____

- A) 1 B) 0 C) 2 D) 11

78. $A = \{a \in \mathbb{Z}^- : y + x = 0 \text{ bisects two chords drawn from a point } \left(\frac{1+\sqrt{2}a}{2}, \frac{1-\sqrt{2}a}{2}\right) \text{ to the circle whose equation is } 2x^2 + 2y^2 - (1+\sqrt{2}a)x - (1-\sqrt{2}a)y = 0\}$

$B = \{K \in \mathbb{Z}^+; (1,4) \text{ lies inside the circle } S \text{ whose equation is } x^2 + y^2 - 6x - 10y + k = 0, \text{ and } S \text{ neither touches the axes nor cuts them}\}$

The sum of max value in set A and minimum Value on set B is

- A) 30 B) 33 C) 24 D) 23

79. Let $S = \left\{ \theta \in \left(0, \frac{\pi}{2}\right) : \sum_{m=1}^9 \sec\left(\theta + (m-1)\frac{\pi}{6}\right) \sec\left(\theta + \frac{m\pi}{6}\right) = -\frac{8}{\sqrt{3}} \right\}$, then which of the following is CORRECT?

- A) $S = \left\{ \frac{\pi}{12} \right\}$ B) $S = \left\{ \frac{2\pi}{3} \right\}$ C) $\sum_{\theta \in S} \theta = \frac{\pi}{2}$ D) $\sum_{\theta \in S} \theta = \frac{3\pi}{4}$

80. $\lim_{x \rightarrow 0} \frac{\tan x \cdot \cos(\sin x) - \sin(\sin x)}{\cos(\tan x) \cdot \tan x - \sin(\tan x)}$ is

- A) $-\frac{1}{2}$ B) -1 C) $\frac{1}{2}$ D) 1

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 10 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Mark nearest Integer only. Have to Answer any 5 only out of 10 questions** and question will be evaluated according to the following marking scheme:

Marking scheme: +4 for correct answer, -1 in all other cases.

81. Given $a, b \in \{0, 1, 2, 3, \dots, 19, 20\}$, consider the system of equations

$x + y + z = 4, 2x + y + 3z = 6, x + 2y + az = b$. Let 'P' denotes number of ordered pairs (a, b) so that system of equations has a unique solution, 'Q' denotes number of ordered pairs (a, b) so that system of equations has no solution and 'R' denotes number of ordered pairs (a, b) so that system of equation has infinite solutions, then

(P+Q+R) is _____

82. The number of divisors of $2^2 \cdot 3^3 \cdot 5^3 \cdot 7^5$ of the form $2n + 1$, $n \in \mathbb{N}$ is _____
83. If the variance of the following frequency distribution is 50,
- | | | | | |
|-----------|---|-------|-------|-------|
| Class | : | 10-20 | 20-30 | 30-40 |
| Frequency | : | 2 | x | 2 |
- then 'x' is equal to _____
84. A particle is moving on a straight line obeying velocity $V = 24t^2(1-t)$. The time 't' when it gets maximum acceleration, is $\frac{1}{\lambda}$ sec, then $\lambda =$ _____
85. $\vec{a}, \vec{b}, \vec{c}$ are the coterminous edges of cube of unit length and $\vec{r}$ be any unit vector inside the cube then $|\vec{a} \times \vec{r}|^2 + |\vec{b} \times \vec{r}|^2 + |\vec{c} \times \vec{r}|^2$ is _____
86. The shortest distance between the lines $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ and $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ is equal to _____.
87. Value of $\sum_{n=1}^{\infty} \tan^{-1}\left(\frac{4n}{n^4 - 2n^2 + 2}\right)$ is $\frac{k\pi}{4}$ then $k =$ _____
88. $\int \frac{3+2\cos x}{(2+3\cos x)^2} dx$ is equal to $\left(\frac{a \sin x}{3\cos x + b}\right) + c$ then $a + b =$ _____
89. The integral $\int_{\pi/6}^{\pi/3} \tan^3 x \cdot \sin^2 3x (2 \sec^2 x \cdot \sin^2 3x + 3 \tan x \cdot \sin 6x) dx$ is equal to $-\frac{1}{k}$ then $k =$ _____
90. let $a = \sum_{r=1}^{11} \sin^2\left(\frac{r\pi}{24}\right)$; $b = \sum_{r=1}^{11} \cos^2\left(\frac{r\pi}{24}\right)$; $c = \sum_{r=1}^{11} \tan^3\left(\frac{r\pi}{24}\right)$ and $d = \sum_{r=1}^{11} \cot^3\left(\frac{r\pi}{24}\right)$ then the value of $a^3 + b^3 - 11(a+b) + 33ab + \log_{(2d-c)}(2c-d)$ is _____